UK Road Safety Analysis:

Accident Trends, Risk Factors & Safety Insights

Active

Updated: Apr 29, 2026

Using 1.8M+ records to identify accident hotspots, severity drivers, and targeted interventions for smarter public infrastructure decisions.

Records	Severity	Risk	Fatal
1.8M+	**1.2%**	**13.2%**	**10,977**
Total accidents	Fatal rate	Serious %	Total fatalities

Dashboard Overview

3 Views

Safety Risk Overview	Temporal Hotspots	Environmental Risk
92%	24	15%
Overall safety score	High-risk hours	Weather impact

Team Members

- **Ved Pawar** — ETL & Analysis
- **Aaryan Krishna** — Dataset & Sourcing
- **Dev Kothari** — EDA & Statistics
- **Yash Agrawal** — Tableau Dashboard
- **Adityaraj Pal** — Report Writing
- **Deepanshu Choudhary** — Presentation

Mentor & Resources

- Archit Raj (Faculty Mentor)
- GitHub: SectionE_G-4_UK-Road-Safety
- Tableau: UKAccidentAnalysis2

Context & Problem Statement

Why UK Road Safety Matters

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Road safety remains a preventable economic and human-cost challenge

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Despite strict traffic regulations, the UK continues to experience a high number of road accidents. 1.8M+ records analyzed to uncover patterns and drive targeted interventions.

1 ! Critical

Challenge

High accident burden persists despite strict regulations. Severe incidents continue.

- 1.2% fatal rate

2 Key Players

Stakeholders

Transport authorities, regulators, urban planners, emergency services.

- 4 key groups

3 Analysis

Problem

Lack of targeted visibility into where and why severe accidents occur.

- 30+ variables

4 Goal

Objective

Turn historical data into actionable insights for targeted interventions.

- Data-driven

Key Insight: Targeted interventions can reduce severe accidents by 15-20%

Analysis comple

Data Engineering Pipeline: Reliability creates trustworthy decisions

Python | Pandas | Tableau

Complete ETL workflow from raw data extraction to interactive Tableau dashboards, ensuring data quality and actionable insights.

1 ✓ Complete
Raw Data
Kaggle UK Dataset
~1.8M rows, 30+ cols
● Multi-year data

→

2 ✓ Complete
Cleaning
Nulls & Duplicates
Missing values treated
● 99.2% clean

→

3 ✓ Complete
Feature Eng.
New attributes
Time buckets, labels
● 15+ features

→

4 ✓ Complete
Validation
Quality checks
No duplicate IDs
● 100% valid

→

5 ✓ Complete
Analysis
EDA & Stats
ANOVA, Chi-square
● 4 tests

→

6 ✓ Complete
Tableau
Dashboards
3 interactive views
● Live

Data Quality Comparison

Before vs After

		Before	After
Missing Values	-95%	9,871	0
Duplicates	-100%	14	0
Invalid Types	-98%	75,133	1,502
Consistency	+99%	85%	99.8%

Tools Stack

Python | Pandas | NumPy | Jupyter | Tableau

Business Impact: Reliable inputs create trustworthy decisions - 99.8% data quality achieved

Processing time: 45 minutes

KPI Framework: Focus on reducing harm, not just counts

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Key performance indicators for monitoring road safety performance, identifying risk patterns, and prioritizing corrective interventions.

Volume	Risk	Severity	Efficiency	Speed	Trend
Total Accidents	**Severe Rate**	**Fatal Count**	**Casualties/Accident**	**High-Speed Fatal**	**Yearly Fatal %**
Volume	Risk	Severity	Efficiency	Speed	Trend
927,320 ↑ +2.3%	**13.2%** ↓ -1.2%	**10,977** ↓ -8.5%	**1.34** ↑ +0.1	**2.93%** ↑ +0.5%	**1.2%** — Stable
Overall safety burden	High-risk share	Loss-of-life impact	Avg human impact	Speed amplifies lethality	Persistent risk
count(Accident Index)	(Serious+Fatal)/Total	count Severity=Fatal	Total/Total Accidents	At 70 mph zones	Range: 1.00-1.42%

Night Share: 18.5% | Weather Risk: 12.3% | Peak Hour: 23.7% | Urban Share: 67.2%

All metrics updated in real-time

Key Insight: Targeted interventions can reduce severe accidents by 15-20%

Last updated: 2 minutes ago

Key Insights (EDA)
What the Data Says

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Volume is urban; severity is environmental and speed-driven

1.2% Fatal

Fatal: 10,977 (1.2% of total)
Fatal incidents drive disproportionate human/economic cost. Total accidents: **927,320** with **10,977** fatalities.

4.03% Night

Night fatal rate: 4.03% vs 0.89% day
Poor visibility drastically increases severity. Unlit darkness vs lit daylight conditions.

2x Rural

Rural ~2% vs Urban ~1% fatal rate
Urban areas drive volume, but rural high-speed roads are structurally riskier.

Severity Distribution

Slight (85.6%)	794,114
Serious (13.2%)	122,229
Fatal (1.2%)	10,977

Source: Safety Risk Overview Dashboard

Long-term Trend
Declining

Volumes decline, but risk remains persistent

Weather Risk Multiplier

Fine	0.89%
Rain	1.21%
Fog/Snow	1.93%

Fatal rate by weather condition

Multi-Vehicle Escalation

x N → x N^2

Positive Relationship Identified
Multi-vehicle crashes drastically raise casualty multipliers

RECOMMENDATION: Prioritize lighting upgrades and implement weather-responsive speed controls in identified high-speed rural corridors to immediately cut fatality rates.

Advanced Statistical Analysis
Evidence Behind the Insights

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Statistical Rigor: Visibility, weather, and speed are significant drivers of severity

α = 0.05

99.9% CI

Hypothesis testing confirms that accident severity varies significantly across environmental and temporal factors, enabling targeted interventions.

ANOVA
Weather

✓ Sig

F-stat:	58.21
p-value:	<0.0001

Accident severity varies significantly across weather conditions

ANOVA
Light

✓ Sig

F-stat:	74.35
p-value:	<0.0001

Lighting conditions strongly influence accident severity

Chi-Square
Urban/Rural

✓ Sig

χ^2:	112.4
p-value:	<0.0001

Significant relationship between area type and severity

Spearman
Vehicles

✓ Sig

r:	0.62
p-value:	<0.0001

Moderate positive correlation between vehicles and casualties

Root Cause Drivers

Poor Visibility
Night, unlit segments → delayed hazard detection

Adverse Weather
Rain/Fog/Snow → longer braking, loss of control

High-Speed Rural
Higher speeds → increased fatality probability

Multi-Vehicle
Chain collisions → higher injury counts

Operational Meaning

Actionable Insights

Light where it matters: Focus on unlit rural corridors
Slow where it kills: Speed enforcement in high-risk zones
Adapt to weather: Real-time traffic alerts during adverse conditions
Manage congestion: Reduce chain collision scenarios

Tableau Dashboard Walkthrough
From Insight to Decision

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Three purpose-built dashboards enable fast drill-downs from KPIs to root causes

Year | Severity | Area Type

Interactive Tableau dashboards provide comprehensive views of road accident patterns, severity drivers, and risk hotspots.

Safety Risk Overview
KPIs & Trends

✓ Active

927,320
Total Accidents

1.2%
Fatal Rate

85%

Severe: 14.4%
Serious: 13.2%

Updated: 2 min ago
View Details →

Temporal Hotspot
Time Patterns

! High Risk

Peak Hours
7-9 AM, 5-7 PM

23.7%
of total

Weekly pattern
Analyze →

Environmental Risk
Weather & Light

✓ Stable

1.93%
Fog/Snow Fatal

+12%
vs baseline

75%

Rain: 1.21%
Fine: 0.89%

Insights →

How Decision-Makers Use This

Policy Leaders
Monitor severe/fatal risk and high-speed corridors

City Ops
Staff patrols by peak hours and days

Infra Planners
Prioritize lighting, weather-alert zones

Tableau Public

public.tableau.com/app/profile/dev.kothari

90% uptime

Open Dashboard

Recommendations

Targeted Actions with Measurable Impact

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Prioritize interventions where risk concentrates — visibility, weather, speed, congestion, hotspots

✓ Ready

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Five data-driven recommendations with implementation roadmap, expected impact, and resource requirements.

Action Matrix

Filter Sort

PRIORITY	ACTION	IMPACT	EFFORT	TIMELINE
↑ High	**Weather-responsive traffic systems** Real-time alerts, adaptive speed limits	15% Severe ↓	✓ Med	3-6 mo
↑ High	**Nighttime safety improvements** Street lighting, reflective signs	12% Fatal ↓	✓ Med	6-12 mo
→ Med	**Rural high-speed enforcement** Speed monitoring, emergency response	18% Rural ↓	Med	3-9 mo
→ Med	**Congestion control strategies** Signal optimization, lane discipline	20% Chain ↓	High	6-12 mo
↑ High	**Hotspot-based targeting** Dashboard-driven interventions	10% Accidents ↓	✓ Low	1-3 mo

Quick Wins | Strategic Bets | Long-Term Build | Key Insight

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Targeted, data-driven interventions deliver material safety gains

✓ On Track

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ROI analysis showing quantified impact across safety metrics, cost savings, and resource optimization.

↑ +15%

10-15%

Severe Accident Reduction

Targeted interventions

↑ +12%

8-12%

Fatal Accident Reduction

Resource optimization

↑ +18%

12-18%

Rural Fatality Reduction

Emergency response

↑ +20%

10-20%

Chain Collision Reduction

Traffic optimization

Cost of Inaction

Fatal Incidents

Persistent severe accidents despite static budgets

Healthcare Burden

Under-served rural corridors drive fatalities

Economic Loss

Missed quick wins from hotspot

Response Delay

Delayed emergency response times

Implementation Timeline

3 Months	Hotspot dashboard rollout, R5 pilots
6 Months	R1 weather alerts, lighting upgrades
12 Months	Scale R2/R4, integrated response

Limitations & Next Steps

From Constraints to Roadmap

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Close critical data gaps now ; build predictive, real-time, smart-city capabilities next.

Ready Now | Medium Term

Address current limitations while building toward advanced analytics capabilities and smart city integration.

Current Constraints

No Driver Behavior Data — CRITICAL
Missing data on distraction, fatigue, alcohol influence

Missing Real-Time Traffic — MEDIUM
No live traffic flow, incidents, response times

Correlation ≠ Causation — ANALYSIS
Unobserved confounders may affect results

Reporting Bias — DATA QUALITY
Police-reported incidents may underrepresent

Future Roadmap

Phase 1: Predictive Model — Ready Now
0-3 months
Severity risk model | KPI governance

Phase 2: Real-Time Dashboard — Medium Term
3-6 months
Stream weather feeds | Alerting system

Phase 3: Smart City Integration — Strategic
6-12 months
Adaptive signals | EMS integration

Team Contributions

End-to-end pipeline delivered with shared ownership

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Thank You for your attention

We appreciate your time reviewing our UK Road Safety Analysis project. This capstone project demonstrates data-driven insights for smarter public infrastructure decisions.

Ved Pawar — ETL & Cleaning
ETL, Data Cleaning & Analysis

Aaryan Krishna — Dataset & Sourcing
Dataset Sourcing & Management

Dev Kothari — EDA & Stats
EDA & Statistical Analysis

Yash Agrawal — Tableau Dashboard
Tableau Dashboard Development

Adityaraj Pal — Report Writing
Report Writing & Documentation

Deepanshu Choudhary — Presentation
Presentation & PPT Design

Faculty Mentor: Archit Raj
Guided the team through data analysis and visualization

Project Status
Completed

GitHub: github.com/Deepanshu-Nst/SectionE_G-4_UK-Road-Safety

Tableau: UKAccidentAnalysis2